

CAO HUY HUNG

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Summary

I'm a Backend Engineer with hands-on experience in building scalable, high-performance systems and integrating AI solutions into production environments. Skilled in designing and maintaining distributed architectures, with practical expertise in Deep Learning applications including Computer Vision and Generative AI. Passionate about optimizing system performance, processing large-scale data efficiently, and delivering reliable, AI-powered products that create real business value.

Experience

Mebisoft
AI Developer

01/2026 – Present
AI Team (Fresher)

- Project 1: AI Virtual Assistant (Mebieco AI Chatbot)*
- Architected a hybrid RAG + Text-to-SQL pipeline (Vanna.ai) enabling the chatbot to query both unstructured knowledge bases and live databases within a single conversational flow.
 - Optimized vector retrieval on PostgreSQL (pgvector + HNSW) via context-aware chunking, improving answer relevance by reducing irrelevant chunk retrieval.
 - Built a fully async backend (FastAPI + Redis) with Redis-backed chat memory and Function Calling orchestration, cutting average response latency significantly.
 - Integrated token usage and latency monitoring, providing real-time observability across the entire AI pipeline.
- Project 2: Distributed Computer Vision System (Mebieco AI CV)*
- Deployed YOLO v8/v11 for instance segmentation, achieving reliable high-density object counting at up to ~3,000 objects/image.
 - Developed a custom measurement pipeline combining Graph Theory, Skeletonization, and ArUco marker calibration to extract real-world dimensions from images.
 - Decoupled API and inference layers via Azure Service Bus, enabling independent scaling and fault isolation across distributed workers.
 - Boosted inference throughput by switching to ONNX Runtime (FP16) on CUDA with batch processing, reducing per-image processing time.

Tech Stack: FastAPI, Python (Asyncio), Azure OpenAI, YOLO, ONNX Runtime, PostgreSQL (pgvector), Redis, Docker, Azure Service Bus, CUDA

University of Transport Ho Chi Minh City (UTH)
Python Developer – Django

04/2025 – 09/2025
Library Information Center

- Designed and maintained RESTful APIs and microservices for the library management platform, improving system modularity and maintainability.
- Integrated OpenTelemetry for distributed tracing, logging, and metrics — reducing average debugging time across Django services.
- Offloaded heavy operations to Celery + RabbitMQ async task queues, improving platform responsiveness under high load.

Education

Sai Gon University
Bachelor of Information Technology

2021 – 2025
GPA: 7.04/10

Skills

Programming Languages	Python, JavaScript
Backend Frameworks	FastAPI, Django, Flask
AI & ML	Ultralytics, LLMs, RAG Architecture, Text-to-SQL, ONNX, OpenCV, Graph Algorithms
Data & Database	PostgreSQL, MySQL, Redis, SQL Optimization, Azure Blob Storage
Messaging & Queues	Azure Service Bus, RabbitMQ, Celery
Infrastructure	Docker, NVIDIA CUDA, Linux, Git, OpenTelemetry
AI Tools	Cursor, Antigravity, Claude Code, Stitch UI